Through the Rearview Mirror

Excavating the Biographical Roots of Equity-Focused Mathematics Teaching

Sam Seim · they/them

Concurrent Convergent Mixed Methods Autoethnography · n = 1

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Background: The Study and Its Researcher

THE STUDY

This autoethnographic dissertation examines how my lived experiences with technology — across my academic career and professional development — shaped my current equity-focused mathematics teaching practices.

The Dissertation Tracker is both the primary research tool and a core data source: a web-based platform I built to document my own reflective journey in real time.

THE RESEARCHER

Hard sciences training → K–12 mathematics teaching

I came into teaching knowing how to use technology with precision. I had no idea how to use it meaningfully — for someone else, in real time.

Homeschooled through high school with significant learning autonomy. That biographical fact is not background noise. It is data.

I am both researcher and participant: n = 1.

Ellis, Adams & Bochner, 2011; Chang et al., 2014

My Definition of Mixed Methods Research

"Mixed methods research is the systematic integration of qualitative and quantitative methods at the level of research questions, data collection, and analysis, wherein quantitative approaches describe what is occurring and qualitative approaches explain why. Like a complex mathematical function that requires both its algebraic form and plotted values to reveal its full behavior across contexts, MMR generates comprehensive understanding that neither could achieve independently."

THE FUNCTION ANALOGY

Algebraic form alone tells you what a function does in theory. Plotted values alone tell you what happened in one context. You need both. MMR is the same: QUAN describes; QUAL explains — neither is optional.

GROUNDED IN (5 MMR scholars)

Greene (2007) — purposeful mixing; multiple ways of knowing
Tashakkori & Creswell (2007) — collect, analyze, integrate
Johnson, Onwuegbuzie & Turner (2007) — breadth AND depth
Morse & Niehaus (2009) — methods-level integration
- Morgan (2014) — pragmatic value-added

Greene, 2007; Tashakkori & Creswell, 2007; Johnson et al., 2007; Morse & Niehaus, 2009; Morgan, 2014

Methodological Positioning: The Field & Its Gap

THE FIELD

Mathematics education research is historically dominated by quantitative methods — especially in studies on educational technology and student outcomes.

MMR has gained traction, but systematic reviews show it still lags behind other education domains in methodological diversity.

(Truscott et al., 2010; Ross & Onwuegbuzie, 2012)

THE GAP

When MMR appears in mathematics education, it focuses on student achievement outcomes — not teacher experience.

What is missing: studies that center teacher positionality, critical reflection, and the biographical roots of instructional practice as primary objects of inquiry.

THE MOVE

Neighboring fields (teacher education, educational leadership) already utilize MMR designs.

Mathematics education has not caught up.

This study contributes a methodological template that treats teacher biography as data — not as confound.

(Chang et al., 2014; Moerer-Urdahl & Creswell, 2004)

Truscott et al., 2010; Ross & Onwuegbuzie, 2012; Chang et al., 2014; Moerer-Urdahl & Creswell, 2004

MMR Rationale: Complementarity

Selected Rationale: Complementarity
Greene et al. (1989); Johnson & Onwuegbuzie (2004)

WHY COMPLEMENTARITY?

The two strands address different-but-complementary dimensions of the same phenomenon:

QUAN: describes the terrain — maps what patterns exist in the Dissertation Tracker artifact record

QUAL: interprets that terrain — the meaning and significance of those patterns

They are not redundant. They are not sequential. They are in dialogue.

Greene et al., 1989; Johnson & Onwuegbuzie, 2004

WHY NOT ANOTHER RATIONALE?

Triangulation requires independent data sources — but as researcher AND sole participant, I am the same source.

Expansion requires the first strand to sequentially inform the second — but my strands are concurrent; I don't need QUAN to identify my QUAL participant.

Complementarity fits: the strands examine different facets simultaneously, then merge for meta-inference.

MMR Purpose Statement

The purpose of this concurrent convergent mixed methods autoethnography is to examine how my lived experiences with technology across my academic and professional development have shaped my current mathematics teaching practices, particularly my commitment to equity and access in mathematics education.

DESIGN

Concurrent Convergent
QUAN + QUAL collected simultaneously, analyzed independently, merged at interpretation
(Creswell & Plano Clark, 2018)

RATIONALE FOR MMR

Complementarity: each strand examines a different-but-related dimension of the same phenomenon. Neither is sufficient alone.

INTENT

Understand how one teacher's biography shapes their present practice — in service of equity and access in mathematics education.

Creswell & Plano Clark, 2018; Guetterman, Fetters & Creswell, 2015

Research Questions

QUANTITATIVE (Correlational)

How do documented patterns in my technology experiences across my academic and professional development relate to observable shifts in my mathematics teaching practices?

QUALITATIVE (Autoethnography)

Central: What critical incidents in my technology experiences have fundamentally influenced my pedagogical philosophy and classroom practices?

Sub-questions address meaning-making in my technology journey and the role of critical incidents in shaping my commitment to equity and access.

MIXED METHODS (Integration)

To what extent do documented patterns in my technology experiences complement and expand upon narrative accounts of critical moments in my development as an educator focused on equity and access in mathematics?

Figure 1 — Concurrent Convergent Design: The Two Strands

Concurrent Convergent Mixed Methods Procedural Diagram — Autoethnographic Study of Technology Experiences and Equity-Focused Mathematics Teaching

STRAND 1 — QUANTITATIVE (Correlational)

RESEARCH QUESTION

How do documented patterns in my technology experiences across my academic and professional development relate to observable shifts in my mathematics teaching practices?

SAMPLE

n = 1 (researcher as participant). Purposive / autoethnographic. Omaha metro area.

DATA COLLECTION (Concurrent)

Dissertation Tracker entries; written artifacts; academic projects across my career; childhood artifacts when available. Ongoing throughout study period.

DATA ANALYSIS

Constant comparative analysis (Charmaz, 2014). Coding frequency, temporal patterns, categorical relationships.

STRAND 2 — QUALITATIVE (Autoethnography)

RESEARCH QUESTION

What critical incidents in my technology experiences have fundamentally influenced my pedagogical philosophy and classroom practices?

SAMPLE

n = 1 (researcher as participant). Same individual — different analytical lens.

DATA COLLECTION (Concurrent)

Self-interview transcripts; Dissertation Tracker narrative entries; thick description of critical incidents.

DATA ANALYSIS

Descriptive thematic analysis + pattern coding (Saldaña, 2021). Reflexive, iterative.

Both strands converge at the Integration / Joint Display stage — continued on next slide.

Figure 1 — Concurrent Convergent Design: Integration

STRAND 1 — QUANTITATIVE (Correlational)

STRAND 2 — QUALITATIVE (Autoethnography)

INTEGRATION · Convergent Design

MIXED METHODS RESEARCH QUESTION

To what extent do documented patterns in my technology experiences complement and expand upon narrative accounts of critical moments in my development as an educator focused on equity and access in mathematics?

QUAN artifact patterns → WHAT

QUAL autoethnographic findings → WHY

Side-by-side joint display merges both strands to generate meta-inference. Analysis moves iteratively between identifying patterns and interpreting meaning.

Creswell & Plano Clark, 2018; Charmaz, 2014; Saldaña, 2021

Table 1 — Joint Display: Autonomy (1 of 2)

Projected joint display of quantitative artifact patterns and qualitative critical incident findings — Theme 1 of 2. Full display at servellon.net/edps936

THEME	QUALITATIVE Projected Self-Interview Excerpt	QUANTITATIVE Projected Artifact Pattern	INTEGRATION Projected
Autono my	*"By high school, my mom gave me a list of things to have done by Friday at 3pm and largely left how and when up to me. When I heard what my peers' school days actually looked like, I didn't feel weird anymore — I felt lucky. I already knew what it felt like to be trusted with your own learning."*	Homeschool-era artifacts projected to reflect 65–70% self-directed, project-based work relative to externally-structured completion — contrasting markedly with formally-structured academic contexts that follow.	**Convergent with Expansion** Both strands confirm homeschool experience established autonomy as a baseline norm. The qualitative strand expands: the contrast with peers' constrained experiences was itself a critical incident — the moment autonomy became visible as something that could be given or withheld, and therefore worth fighting for in a classroom.

Theme 2: Reflection Through Making — next slide.

Table 1 — Joint Display: Reflection Through Making (2 of 2)

Projected joint display of quantitative artifact patterns and qualitative critical incident findings — Theme 2 of 2. Full display at servellon.net/edps936

THEME	QUALITATIVE Projected Self-Interview Excerpt	QUANTITATIVE Projected Artifact Pattern	INTEGRATION Projected
Reflection Through Making	*"I wrote the most in the tracker when I was in the middle of building something — not when I set aside time to reflect. The tracker didn't just record what I was doing; building it was what made me understand what I was doing."*	Approximately 70–75% of high-frequency documentation windows in the Dissertation Tracker are projected to co-occur with periods of artifact creation or active tool engagement, compared to documentation rates during periods of passive use or reading.	**Convergent and Mutually Constitutive** Both strands confirm making functions as a catalyst for reflection rather than the reverse. This is the methodological core of the study: the QUAN pattern grounds the QUAL claim as methodology rather than preference; the QUAL narrative makes the QUAN pattern interpretable as meaningful rather than incidental. Together, they are the argument.

Note. Full joint display (6 themes) available at servellon.net/edps936. Themes shown selected for centrality to the study's core argument.

Addressing the Most Relevant Validity Threats

Most relevant threats to validity in a concurrent convergent autoethnographic design:

STAGE	THREAT	MY PLAN
Integration During Design	Selecting a convergent design when research questions call for a different structure; choosing a core design misaligned with the study's aims.	Both strands target parallel, complementary concepts — QUAN describes what patterns exist in my documented biography; QUAL explains why those patterns emerged and what they mean. As researcher AND sole participant, I don't need the quantitative strand to identify qualitative participants or design the interview protocol. Convergent is the correct structural fit.
Integration During Interpretation	Findings from the QUAN and QUAL phases are not adequately integrated to generate sound meta-inferences. Divergent findings are left unaddressed.	Autoethnography as both design and method ensures nuanced, reflexive interpretation. Because I am both participant and researcher, I am guaranteed to encounter findings that challenge my own self-concept — these divergent moments are treated not as problems to explain away but as sites of additional analytical cycles. The joint display provides structural transparency for convergence and divergence alike.

Note. Full validity table (5 integration stages) is available in the course portfolio. Threats shown selected as most structurally significant for this autoethnographic concurrent convergent design.

Validity framework: Creswell & Plano Clark, 2018; Seim (MMRValidityTableConvergent)

Congruence, Contribution & So What

Definition	Positioning	Rationale	Purpose Statement	Research Questions	Procedural Diagram	Joint Display	Validity

All elements hang together as a coherent, congruent MMR proposal.

CONTRIBUTION TO THE FIELD

· First concurrent convergent MMR autoethnography in mathematics education centered on teacher biography

· Bridges constructivist + postpositivist epistemologies without privileging either

· Treats teacher positionality as data — not as confound to be controlled for

· Offers a methodological template for equity-focused self-study in math education

THE CORE CLAIM

"I wrote the most in the tracker when I was in the middle of building something — not when I set aside time to reflect. The tracker didn't just record what I was doing; building it was what made me understand what I was doing."

Making is not a precursor to reflection. Making is reflection. That is the methodological heart — and why "reflection through making" names both the method and the finding.

Through the Rearview Mirror

Questions??

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